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Discussion Paper

New aspects for the physiotherapy of pushing behaviour

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1. Introduction

While most patients with hemiparesis have good trunk balance soon after stroke, some patients show the peculiar behavior of using the non-affected arm or leg to push away actively from the non-paralyzed side when standing or sitting (Fig. 1). Davies [1] termed this pathological behaviour the '*pusher syndrome*'. If not prevented, the patients push themselves laterally to the point that they fall toward the hemiparetic side. There is forceful resistance against interventional attempts to correct their tilted posture.

Using magnetic resonance imaging and computed tomography, a recent study investigated the lesion location that is typically associated with pusher syndrome [3]. In 46 patients with and without contraversive pushing, the overlap area of infarction was determined. The analysis revealed that the brain structure typically damaged in patients with pusher syndrome is the left or right posterior thalamus. Recently, also the mechanism leading to contraversive pushing has been investigated. The authors examined the ability of patients with pusher syndrome to determine their own upright position while their eyes were occluded. They found an altered perception of the body's orientation in relation to gravity. Patients with pushing behaviour experienced their body as oriented 'upright' when actually tilted about 20° to the ipsilesional side [4]. Surprisingly, these patients showed undisturbed processing of visual and vestibular inputs determining visual vertical. Thus, in contrast to their disturbed perception of upright body posture, orientation perception of the visual world was unaffected. In other words, patients with pushing behaviour have a severe tilt of body posture, in spite of normal visual-vestibular functioning.

This discrepancy of a pathologically tilted *postural* vertical concurrent with an unimpaired perception of the *visual* vertical shows that patients with pushing behaviour apparently manifest a selective disturbance of the control of upright body posture. While the patients were no longer able to tell when their body is oriented in an erect position, they had, on the other hand, no problems determining the orientation of the visual world around them correctly.

These new insights into the mechanisms leading to pushing behaviour encouraged us to suggest a new approach for the physiotherapy of patients with pushing behaviour [2]. The purpose of this article is to describe in more detail this new approach centering around the patients' preserved ability to align the body axis to earth-vertical with the help of visual cues. Moreover, we present our experience of how the clinical management of patients with pusher syndrome in situations such as transfering them from bed to wheelchair and positioning the patients stablely in the wheelchair might improve.

2. Davies' approach for treating pusher syndrome

Davies [1] described several activities, mainly for specific treatment of the patient's upright position. Pas-

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Fig. 1. A right brain damaged patient with contraversive pushing. The characteristic features of the disorder are that these patients while sitting (left) and standing (right) spread the non-paretic extremities from the body to push away actively from the non-paretic side. The result is the typical tilted body posture. If not assisted by the examiner, the patients push themselves into a lateral inclination until they fall towards the hemiparetic side.

sive movements were suggested to restore free lateral flexion of the head. Flexor activity should be stimulated on the paretic side of the trunk during different activities in sitting and standing. An important goal of the treatment strategy put forward by Davies is that patients learn to place weight on both the non-paretic and the paretic side. For regaining the midline in standing, she recommended the use of a knee extension splint to support the paretic leg. With such a splint, the standing patient is able to carry out activities with minimal help from the therapist. To elongate the non-paretic side and avoid pushing behavior while standing, activities with the non-paretic extremities are carried out. A plinth on the sound side or in front of the patient is used for orientation. In a subsequent phase of the treatment, the therapist removes the extension splint and assists the patient in walking. Patients who are still unable to stand and walk often manage to balance their weight on climbing stairs. Davies thus recommended bringing patients with contraversive pushing in an upright position as soon as possible and assisting them to walk and climb stairs.

3. New suggestions for the physiotherapy of pusher syndrome

3.1. Perception of the tilted body position and visual exploration

The recent finding that patients with pushing behaviour suffer from impaired perception of the body's orientation in relation to gravity [4] prompts to treat these patients in an earth-vertical position, i.e. while sitting, standing, or walking. According to our clinical experience, it is initially helpful to demonstrate to the patients their falling from extreme lateral inclinations by allowing pathological pushing to occur. The subsequent fall then is arrested while making the patients visually aware of their tilted body position. As soon as they notice their tilt, they are given assistance for active correction. This includes support with the non-paretic hand on the paretic side, or a shift of the non-paretic arm and thus the center of gravity towards the non-paretic side. The experience of consequently not falling in this situation can be used for positive reinforcement. Moreover, since orientation perception of the *visual* surroundings is *not* impaired in patients with contraversive pushing [4], they can see that they are not in an erect position by looking at their structured surroundings. However, they cannot spontaneously make use of this preserved ability; they have to be trained to do so. Since the patients *feel* erect when they *see* that they are tilted and vice versa [4], the first goal of physiotherapy should be to demonstrate this, showing the patients that visual information corresponds to reality. While sitting or standing, the patients should be asked to *see* whether they are oriented upright. Therefore, it is useful to work in a room containing vertical structures, such as door frames, windows, pillars, or pictures. It is also beneficial to use visual aids that give the patients feedback about their body orientation (Fig. 2). It is our

Fig. 2. Patient with left-sided hemiparesis and with contraversive pushing (left). Since orientation perception of the visual surroundings is not impaired in these patients [4], they realize they are not in an erect position by looking at their structured surroundings and can align their body with these earth-vertical structures. In addition, the therapist may give visual aids – such as using the arm shown here – to demonstrate the earth-vertical, upright orientation (right). With little help, the patient is now able to orient his body upright.

clinical observation that the experience of not falling after attaining the corrected position, combined with seeing that they are upright, increases the patients confidence, and lowers the reaction to abduct and extend non-paretic extremities.

3.2. Reaching a vertical body position actively

Whereas passive movements are employed in the diagnosis of contraversive pushing, such movements do not appear to be useful in therapy. Patients with contraversive pushing respond to passive change in body orientation with active resistance. However, if they are shown an object at their non-paretic side (e.g., the therapist's hand) and are asked to reach the object with the hand by shifting their weight toward this side, they are able to desist from pathological pushing temporarely (Fig. 3). To assist the patient, the therapist may demonstrate the movement requested. Acoustic signals, such as knocking on the bed frame or the therapy table, can help to show the patient the object to be reached. If the therapist is sitting on the non-paretic side of the patient, the therapist him/herself may serve as such 'object'.

3.3. Transfer from bed to wheelchair

The transfer from bed to wheelchair is especially problematic with patients showing pushing behaviour in addition to a hemiparesis. It must be prepared carefully to prevent the patients from falling and to minimize the therapist's effort.

It is our experience that the patients first should lose their fear of bending forward. They are asked to stroke their thighs from hip to knee, with both hands. If they succeed without fear, we ask them to stroke further to their lower legs and back to their thighs. The therapist guides the paretic hand. Thus the patients automatically shift their weight forward, toward their feet. The *low transfer* corresponds to the normal movement of changing from one seat to another close by. The opportunity to feel, with one hand at least, where they are coming from and where they are going to, seems to increase the patients confidence and independence. The buttocks are lifted just enough to pass over the tires of the wheelchair. The patients then have to master a small swinging movement of the buttocks to change seats. The patients lean far forward, if possible supporting themselves with one hand next to the buttocks and the other on the side of the wheelchair. If the paretic hand cannot be used, it rests between the patient's thighs. The therapist stands in front of the patient and directs the patient's legs from the outside guiding with her feet and knees. In this way the therapist also keeps the non-paretic leg from abducing, to prevent its use to laterally push during the transfer. The therapist's hands should grasp the patient from above at the outer scapular borders, near the axillae.

We made the experience that, at the beginning, it is easier to transfer *toward* the paretic side, since patients

Fig. 3. Patient with left-sided hemiparesis and with contraversive pushing. The patient is shown a goal to be reached at the non-paretic side (here the therapist's hand). The patient now is able to desist from pathological pushing and can shift weight toward this side.

tend to push in this direction. On transferring via the non-paretic side, patients tend to press contrary to the direction desired. Hence time must be spent teaching them to support themselves with their hand instead of pushing away. Initially nothing can be said against transferring with the patients in the easiest way, until they learn to deal better with their problem of perceiving upright body posture. Later on, the transfer should be practiced in both directions.

3.4. Positioning in the wheelchair

As illustrated in Fig. 4, the patients should sit with their buttocks at the very back and in the middle of the wheelchair. On the paretic side, a cushion between the scapula and the back rest impedes levering the buttocks toward the non-paretic side. The arms should rest in front of the patients on an attached table. The feet should both be placed on the ground.

Even if all these measures are taken, patients tend not to sit like this for long. They push their shoulders toward the paretic side and the back, so that their buttocks slide forward toward the non-paretic side. The table board does not prevent them from falling. In order to return to the correct position, we tell the patients to lean forward and slide their buttocks back. A therapist assists them, holding the outer scapular borders from above, while facing the patient. An additional assistant may be necessary to move the buttocks toward the back.

4. Conclusions

Contraversive pushing is a disorder that obviously has a good prognosis. Only 6 months after a stroke, pathological pushing was rarely still evident in an unselected sample of patients with initially severe pushing behaviour [6]. But we also know that patients with contraversive pushing take 3.6 weeks (= 63%) longer than patients without that disorder, to reach the same functional outcome level [7]. Physiotherapy of contraversive pushing thus aims to shorten this period. Patients with contraversive pushing should become inde-

Fig. 4. Patient with left-sided hemiparesis and contraversive pushing sitting in a wheelchair. Spontaneously, the patient is pushing with the non-paretic arm toward the hemiparetic side leading to severe tilted body posture (left). To stabilize upright position (right), the patient sits with the buttocks at the very back. On the paretic side, a cushion between the scapula and the back rest impedes the patient's levering the buttocks towards the non-paretic side. The arms rest in front of the patient on an attached desk.

pendent of help from other persons in less time and be discharged from inpatient care earlier.

Although patients with contraversive pushing appear to have an altered perception of the body's orientation in relation to gravity, they are not impaired in perceiving upright orientations in the surroundings *visually* [4]. However, they are not able to make use of this ability spontaneously to control upright body posture. This seems to become possible when training procedures apply this ability for conscious strategies of posture control. We made the clinical experience that the treatment plan is most effective if designed in such a way that the patients learn to realize the disturbed perception of erect body position, visually explore the surroundings and the body's relation to the surroundings, use visual aids that give feedback about body orientation, learn the movements necessary to reach a vertical body position, and, finally, learn to maintain the vertical body position while performing other activities.

First systematic observations slowed that the "Visual Feedback Training (VFT)" leads to significant improvement [8] but we need further controlled studies to evaluate whether or not these suggestions for the physiotherapy of patients with pusher syndrome are effective to shorten the time for inpatient care and to accelerate independence in daily linving. An important aspect for the comparability of such studies will be that the diagnosis of pusher syndrome becomes more standardized in clinical practice and also allows to quantify the severity of pushing behaviour. Karnath and Broetz [2] recently made a suggestion of how such a procedure could look like. A helpful tool in this context also might be the standardized 'Scale for Contraversive Pushing (SCP)' [4,5]. The SCP assesses (a) spontaneous body posture, (b) increase of pushing force by spreading of the non-paretic extremities from the body, and (c) resistance to passive correction of posture. In conclusion, we hope that this article will help to stimulate the efforts needed to search for further improvement of the physiotherapy of pusher syndrome.

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